Developed through the [BiTS] program, a part of [Renaissance Philanthropy]

VAPOR

Velocity **A**ttenuation via **P**orous **O**rbital **R**eefs

Imagining new protection for the Space Economy
As orbital usage expands, the escalating density of space debris poses an existential threat to future infrastructure; VAPOR aims to secure the viability of Low Earth Orbit (LEO) by developing space-deployable porous material shield architectures designed reduce impact forces and extend the service life of critical space assets. This effort is intended to be a high-impact ARPA-like program and will

The Trajectory of Orbital Risk

The historical trajectory of space debris began as a byproduct of a "linear" space economy, and cradle-to-grave thinking covered launch to atmospheric burnup. But statistically speaking, every object launched now generates roughly 10,000 fragments of debris—and has now reached a critical inflection point. For decades, the vastness of the "cosmic ocean" allowed for a philosophy of neglect, but with the acceleration of the commercial space race and massive satellite “megaconstellations” increasing LEO object density by at least 7x, the probability of collisions has increased exponentially. We are moving from an era of isolated incidents to a period of "orbital congestion," where the sheer volume of untracked micro-objects threatens the very satellites that power our global economy, defense, and communications. Governments are taking notice, and policymakers are focused on one thing: reducing harm. VAPOR wants to make space protection a benefit by changing the trajectory of space debris formation.

The Cost of Inaction and the Current Bottleneck

If we do not innovate, we face mitigating for the "Kessler Syndrome": a runaway chain reaction of collisions where the risk of operating in space eventually outweighs the benefits, potentially rendering specific orbital planes unusable for generations. The primary bottleneck keeping us on this path is the limitation of current protection methods. Currently, operators must choose between "dodging" which consumes finite fuel and requires perfect foresight—or traditional "hard armor," which is mass-prohibitive for modern launch vehicles and often creates more debris upon impact. We lack a material solution that is lightweight, scalable, and capable of mitigating damage without contributing to the debris field.

A New Horizon of Opportunity

If we shift our approach from rigid resistance to velocity attenuation, we unlock a new paradigm of space stewardship. By deploying "orbital reefs," we transform the protection of individual assets into a collective improvement of the orbital environment. This self-filtering effect offers the opportunity to not only shield expensive, long-lived infrastructure but also to potentially capture incumbent debris, turning a passive defense into an active cleaning mechanism. This transition secures the current $630B global space economy and provides a pathway to >$10T.

More importantly, it paves the way for a sustainable, circular economy in orbit, ensuring that the shores of our cosmic ocean remain open for the next hundred years of human exploration and development.

Program Thesis

Developing new materials that will absorb the impacts of incumbent space debris while also preventing the creation of new debris will change the trajectory of space management practices before a critical disaster emerges. The VAPOR program is predicated on the fundamental shift from traditional "Whipple shields" and hard-plate armor toward gradient, porous architectures that utilize controlled failure modes to manage hypervelocity (>3 km/s) impacts. At these orbital velocities, impacts often result in the fragmentation of both the projectile and the shield, contributing in a reinforcing feedback loop to the "cloud" of deleterious micro-debris.

VAPOR proposes a "ballistic gel" approach, where specially engineered porous structures slow projectiles through successive, high-surface-area interactions, capturing the debris and potentially allowing for healing opportunities. Material discovery may be aided through AI-enhanced approaches Like DiffSyn or Lila.ai to navigate the huge variety of combinations of materials and increase the speed at which scientific iteration is possible. New design tools like H-MAD have also come out to assist in the design of hierarchical materials—types of materials that are specially arranged across multiple length scales (nm, mm, m, etc.) to generate properties that are not possible in traditional solid bulk materials. VAPOR aims to leverage the capability of engineered mechanical properties to reduce weight through the introduction of purposeful voids. The intention is that any impactor will encounter a material that can undergo significant deformation, losing energy across a wide area as the porous medium stretches and flexes. At technological maturity, it is possible to envision a flowing system of supportive nutrients to enable VAPOR armor to self-heal and extend operational flight time.

This strategy aims to provide superior protection for the next generation of space infrastructure, including commercial space stations and large-scale orbital compute nodes. Beyond this, there is accelerating interest in using space for manufacturing to make products that are impossible or prohibitively expensive to manufacture on Earth. This is due to the unique opportunities afforded by both a microgravity environment and a high vacuum environment. As new industries seek to use space, it becomes increasingly important to ensure permanent access.

Strategically, the VAPOR thesis integrates the necessity of "space heritage." The program is structured to move rapidly from chemical and composite formulation to in-space demonstrations. This is essential because the space environment remains the ultimate proving ground for material longevity. By leveraging relationships with commercial launch integrators, VAPOR will provide the tangible evidence required for these unproven materials to be adopted by risk-averse mission architects.

The technologies developed out of VAPOR offer the opportunity for regulators and businesses to work together. By switching the incentive structure from a cutthroat environment to one where cooperation and trust are rewarded, the VAPOR program can affect space infrastructure coordination on a global scale.

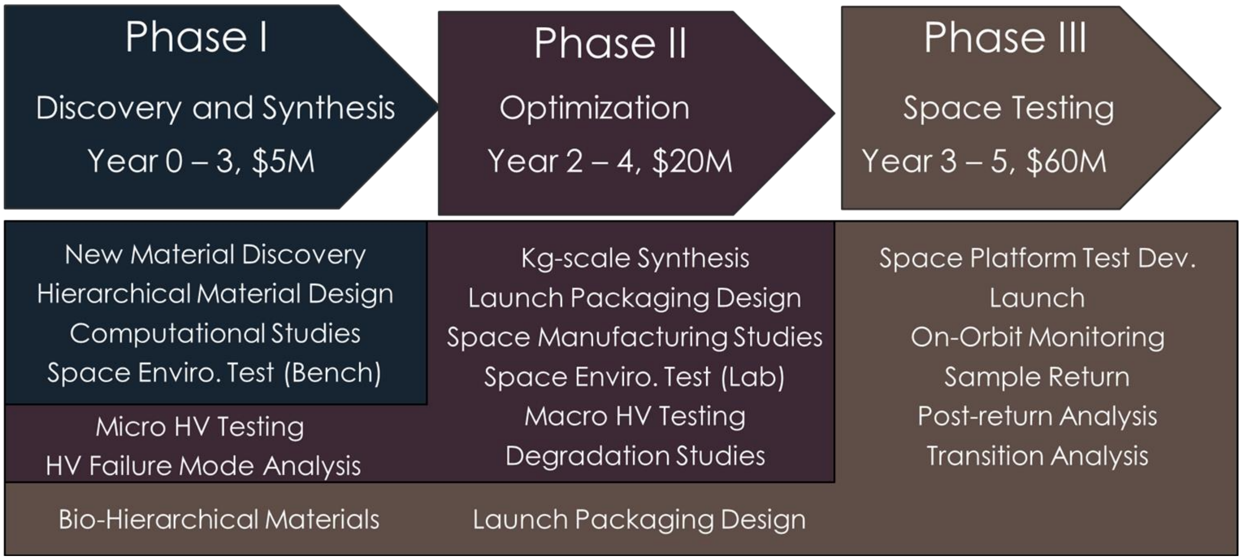

Figure 1: Proposed phasing structure of VAPOR program. The colored blocks outline activities that are possible at different funding gates; additional funding up front could be allocated for more expensive research in parallel.

Program Resource Allocation

The VAPOR program is designed as a focused, high-impact research initiative appropriate for various Advanced Research agencies or philanthropic consortia with a total projected budget of $60M. This investment is strategically divided between three program phases to ensure that the discovery of new materials is immediately supported by high-fidelity testing and space-qualification. Many performer teams are possible to be funded under Phase I of this structure, with capability teams supporting the successful candidates from Phase II and beyond.

The program is structured across three distinct phases over a five-year timeline:

1. **Phase 1: Discovery and Synthesis (Years 0-3):** Initial efforts focus on the formulation of new chemical and composite structures, supported by computational fluid dynamics and micro-object hypervelocity testing.
2. **Phase 2: Development and Packaging (Years 2-4):** Research scales to kilogram-level synthesis and centers on launch packaging design. During this stage, the program explores space manufacturing possibilities and conducts longer-duration environmental testing.
3. **Phase 3: Space Testing (Years 3-5):** The final phase involves the actual launch and deployment of test platforms (potentially on the ISS or other commercial destinations). This phase culminates in sample return and a final economic assessment to determine the material's degradation rates and its readiness for commercial transition.

This phased approach allows for funding "gates," where additional capital can be unlocked for parallel research paths if early milestones in velocity reduction and mass efficiency are met. VAPOR is explicitly limited to this early-stage demonstration; the final deliverable is a material ready for higher Technology Readiness Level funding streams, ensuring that the technology can be seamlessly integrated into the burgeoning space economy of the coming decades.